

Kepang AI

Decision Intelligence untuk Resiliensi Ketahanan Pangan Daerah

Combined Submission Attachment - Tahap 2 - Prepared 4 June 2026

Purpose

One PDF package for judges: problem evidence, solution fit, architecture, data policy, impact KPI, business model, MVP readiness, and validation gaps.

Field	Content
Team ID	[ISI ID TIM SESUAI PENDAFTARAN]
Team Name	J4
Proposal Title	Kepang AI: Decision Intelligence untuk Resiliensi Ketahanan Pangan Daerah
Prototype Status	Local web prototype with backend API, Supabase database, source-aware forecasting, logistics recommendation, and Evidence Room
File Naming Rule	Rename final file to: [ID Team] - Kepang AI Decision Intelligence untuk Resiliensi Ketahanan Pangan Daerah.pdf

1. Executive Evidence Summary

Kepang AI is a decision intelligence prototype for food resilience. The core problem is policy lag: price, weather, stock, logistics, and macro pressure are often analyzed separately, so intervention can arrive after food prices have already moved.

The solution connects official data, real-time-capable feeds, model-derived forecasts, and explicit data gaps into one decision cockpit. The app turns signals into Resilience Score, scenario shock, route recommendation, action cards, KPI targets, and data lineage.

Evidence Signal	Why It Matters	Kepang AI Response
Volatile food 6.24% yoy	Food commodities remain a current inflation pressure	Food Inflation Monitor and commodity risk
USD/IDR Rp17,700	Weaker rupiah raises imported inflation and logistics exposure	Rupiah scenario planning and import exposure
Weather risk	Flood, drought, and climate shocks affect production	BMKG-based Weather Risk Engine
Regional imbalance	Surplus and deficit regions need coordinated distribution	Supply-demand map and route recommendation
Data trust gap	Forecast, seed, and official data must not be mixed	Data lineage labels and expected data fields

2. Evidence of Demand

Demand is demonstrated by public problem evidence and institutional need. Stakeholders do not only need another dashboard; they need a shared decision layer that explains what happened, why it matters, what action should be prioritized, and which data can be trusted.

Real Problem Evidence	Source	Demand Signal
Inflasi Mei 2026 mencapai 3.08% yoy dan 0.28% mtm	BPS, 2 June 2026	Price stability is an active public issue
Volatile food mencapai 6.24% yoy	Antara citing BPS	Strategic food commodities still drive inflation pressure
Rupiah Rp17,700/USD and BI-Rate 5.25%	Bank Indonesia RDG May 2026	Macro shock requires resilient scenario planning
Imports Jan-Feb 2026 reached US\$42.09B, +14.44% yoy	BPS export-import release	Imported inputs, energy, and logistics can become more expensive
Bapanas mission includes reserve, logistics, distribution, stabilization, and vulnerable regions	Bapanas Renstra 2025-2029	Solution aligns with institutional mandate
BMKG provides 3-day village-level weather forecast, updated twice daily	BMKG Open Data	Weather data can feed early warning

3. Problem-Solution Mapping

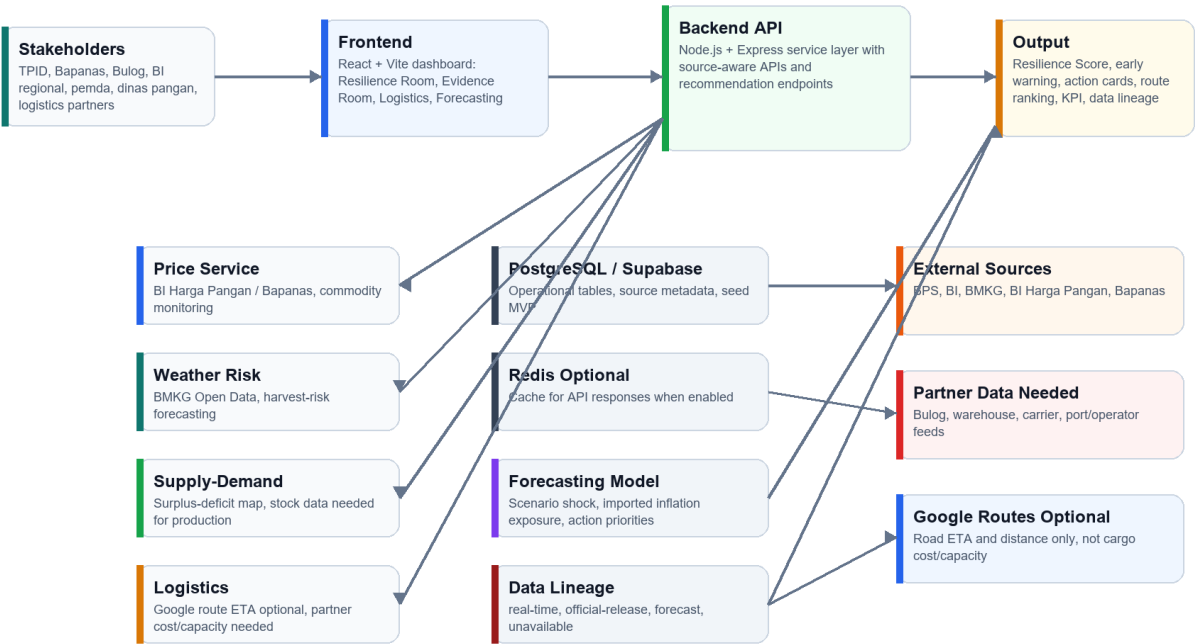
Problem	Mechanism	Outcome
Fragmented food data	Integrate price, supply-demand, weather, macro, and logistics in one cockpit	Decision makers see root causes, not only price symptoms
Policy lag	Early warning, Resilience Score, action cards, redistribution recommendation	Stock intervention, market operation, and harvest mitigation can start earlier
Weaker rupiah and imported inflation	Currency scenario planning, import exposure, logistics-cost simulation	Food decisions become more resilient against external shocks
Unclear data quality	Data lineage: real-time, official-release, forecast, unavailable	Every recommendation is more auditable and honest
Suboptimal logistics route	Route ranking, volume estimate, optional Google ETA/distance, expected partner data	Distribution from surplus to deficit regions becomes more targeted

4. System Architecture

The architecture separates user interface, API orchestration, data storage, external data connectors, model outputs, and data lineage. This supports a transparent MVP today and a production path with institutional data partners later.

Kepang AI System Architecture

Decision cockpit that turns food price, weather, supply-demand, logistics, and macro signals into auditable actions.



5. Data Source and Feasibility Matrix

Dataset	Current Status	Source / Method	Production Requirement
Inflation and macro	official-release	BPS and BI releases	Periodic official update ingestion
Weather forecast	real-time capable	BMKG Open Data	adm4 mapping for all target regions
Food prices	real-time capable	BI Harga Pangan scraper/API; Bapanas when stable	Stable endpoint access and source logging
Supply-demand	forecast / seed MVP	Prototype seed; expected Bapanas/Bulog/Dinas data	production_ton, stock_ton, demand_ton, warehouse_id
Logistics ETA/distance	real-time capable when key exists	Google Routes API for eligible road routes	Google key and road-routable origin/destination
Logistics cost/capacity	forecast / unavailable	Prototype route table	carrier, capacity_ton, cost_per_ton, lead_time_days
Resilience score	forecast	Kepang AI model	Validation against historical interventions

Data honesty rule

Kepang AI must not present seed, proxy, or model output as audited real-time data. Every displayed dataset is labelled as real-time, official-release, forecast, or unavailable.

6. Impact Measurement

KPI	MVP Target	Measurement Method
Time to identify priority region	Under 3 minutes	User task test with TPID/dinas pangan persona
Time-to-insight reduction	50% vs manual baseline	Compare dashboard task vs spreadsheet/manual workflow
Simulated supply gap reduction	At least 30%	Before-after redistribution scenario
Logistics cost potential	5-10%	Cost per ton before-after route optimization
Data lineage completeness	100%	Audit UI and API response
Recommendation validation	5-10 interviews	Structured interview scorecard

7. Business Model and Sustainability

Revenue Stream	Buyer	Value Sold
Annual institutional license	Pemda, TPID, Bapanas, Bulog, BI regional	Dashboard, user seats, modules, regional coverage
Implementation fee	Pilot institution	Setup, data integration, training, custom workflow
Managed analytics	Government or enterprise partner	Monthly risk report and scenario briefing
API subscription	Institutions, logistics, finance, insurance	Risk score, anomaly signal, route intelligence
Logistics add-on	Carrier, warehouse, commodity aggregator	Route optimization, backhaul, capacity matching
Grant / innovation funding	Public program, CSR, donor	Early pilot, validation, and social impact deployment

The product monetizes the intelligence layer, not raw public data. Profitability can come from recurring institutional subscriptions, implementation services, managed analytics, and API/logistics add-ons once pilots prove measurable value.

8. Guidebook Alignment

Guidebook Criterion	Evidence in Kepang AI	Status
Alignment with Problem Statement	Digitalisasi Ketahanan Pangan, policy lag, supply-demand, price, weather, logistics, and rupiah	Ready
Effectiveness & Impact	Resilience Score, action cards, KPI targets, supply gap, route efficiency	Ready for MVP
Uniqueness / Creativity	Decision cockpit, rupiah shock scenario, data lineage, source-aware logistics	Ready
Technical Quality	React/Vite, Express, PostgreSQL/Supabase, BI scraper, BMKG service, Google optional	Prototype ready
Business Model Feasibility	B2G SaaS, implementation fee, managed analytics, API subscription, logistics add-on	Ready

9. MVP Readiness and Test Evidence

Check	Result	Observation
Frontend build	Pass	Vite production build completed; chunk-size warning non-blocking
Backend syntax checks	Pass	forecast route, logistics route, googleRoutesService valid
GET /api/health	Pass	status=ok, database=ok, redis=disabled locally
GET /api/forecast/resilience	Pass	Macro context, resilience score, data policy, data provenance returned
GET /api/logistics/recommendations	Pass	Ranked routes, source policy, expected production data returned
Browser checks	Pass	Evidence Room and Dashboard render with no mojibake

10. Open Gaps and Next Validation

- Fill final Team ID and confirm official team composition before submission.
- Run 5-10 structured interviews with TPID, dinas pangan, Bulog/logistics, BI regional, commodity traders, and warehouse/carrier operators.
- Integrate production stock, warehouse, carrier capacity, port congestion, and actual cost-per-ton feeds when partner access is available.
- Configure Google Maps keys only for road ETA/distance; do not claim sea/cargo cost or capacity as Google real-time data.
- Prepare a 2-3 minute demo video as backup if public hosting is deferred.

11. Source Links

Source	URL
BPS Inflasi Mei 2026	https://www.bps.go.id/id/pressrelease/2026/06/02/2579/inflasi-year-on-year--y-on-y--pada-mei-2026-sebesar-3-08-persen-.html
Antara/BPS volatile food	https://www.antaranews.com/berita/5590715/bps-kenaikan-bbm-nonsubsidi-dorong-inflasi-transportasi-061-persen
BI RDG Mei 2026	https://www.bi.go.id/id/publikasi/ruang-media/news-release/Pages/sp_2810726.aspx
BPS Ekspor-Impor Februari 2026	https://www.bps.go.id/assets/pressrelease/2026/04/01/2557/ekspor-dan-impor-indonesia-februari-2026-masing-masing-tercatat-usd-22-17-miliar-dan-usd-20-89-miliar-.html
BPS NTP Mei 2026	https://www.bps.go.id/id/pressrelease/2026/06/02/2580/nilai-tukar-petani--ntp--mei-2026-sebesar-127-73-atau-naik-1-99-persen.html
BMKG Open Data	https://data.bmkg.go.id/prakiraan-cuaca
Bapanas Renstra 2025-2029	https://peraturan.go.id/filespengundangan/peraturan-bapanas-no-9-tahun-2025.pdf